

Question:

Two walls and the ceiling of a room meet at right angles at point \$P\$. A fly is in the air one meter from one wall, eight meters from the other wall, and nine meters from point \$P\$. How many meters is the fly from the ceiling?

(A) $\sqrt{13}$ (B) $\sqrt{14}$ (C) $\sqrt{15}$ (D) 4
(E) $\sqrt{17}$

Reasoning + Answer:

<think>

1. Identify the coordinates and distances: We are given the following information about the fly :

- The fly is one meter from one wall .
- The fly is eight meters from another wall .
- The fly is nine meters from point (P) .

2. Visualize the room and the fly: Let 's place the room in a 3 D coordinate system . Assume :

- Point (P) is at the origin $((0 , 0 , 0))$.
- One wall is along the (x) -axis (i.e., along the line $(y = 0)$ and $(z = 0)$).
- Another wall is along the (y) -axis (i.e., along the line $(x = 0)$ and $(z = 0)$).

3. Set up the fly 's position: Let the fly be at coordinates $((x , y , z))$. Given :

- $(x = 1)$ (one meter from one wall).
- $(y = 8)$ (eight meters from the other wall).

4. Use the distance from point (P) : The distance from the fly to point (P) is 9 meters . Using the distance formula :

$$\sqrt{(x - 0)^2 + (y - 0)^2 + (z - 0)^2} = 9$$

Simplifying , we get :

$$\sqrt{x^2 + y^2 + z^2} = 9$$

Squaring both sides :

$$x^2 + y^2 + z^2 = 81$$

5. Substitute the known values: We know $(x = 1)$ and $(y = 8)$. Substitute these into the equation :

$$1^2 + 8^2 + z^2 = 81$$

$$1 + 64 + z^2 = 81$$

$$65 + z^2 = 81$$

$$z^2 = 81 - 65$$

$$z^2 = 16$$

$$z = \sqrt{16} = 4$$

6. Conclusion: The fly is 4 meters from the ceiling .

< answer > 4</answer>